

SOC FLAGSHIP INVESTIGATION REPORT

Cross-Platform Threat Detection in Microsoft Sentinel

Two confirmed investigations spanning Linux credential access (MITRE T1110) and Windows execution (MITRE T1059.001), built on a self-deployed Microsoft Sentinel detection pipeline.

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Platform: Microsoft Sentinel (Unified Defender Portal)

Workspace: law-soc-lab, Azure East US

Investigations: SSH Brute-Force (T1110), Malicious PowerShell (T1059.001)

Repository: github.com/WiLL75G/sentinel-soc-lab-setup

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1. EXECUTIVE SUMMARY

This report consolidates two end-to-end security investigations conducted on a Microsoft Sentinel detection environment built from the ground up. Beginning from an empty Azure Log Analytics workspace, a full telemetry pipeline was engineered, detection logic was authored in KQL, two distinct attacks were simulated and hunted, and both were confirmed and contained.

The first investigation addressed an SSH brute-force attack against a Linux host. A single source IP produced 88 failed authentications and 8 successful logins against one account, a confirmed credential compromise, contained at the host firewall. The second addressed malicious PowerShell on a Windows endpoint, where encoded commands, a download cradle, and an execution-policy bypass were detected through script block logging, defeating base64 obfuscation.

Together these investigations demonstrate cross-platform detection capability across two operating systems, two log sources, and two MITRE ATT&CK tactics, Credential Access and Execution, alongside the pipeline engineering, threat hunting, and incident documentation expected of a Tier 1 SOC analyst.

2. ENVIRONMENT AND DETECTION PIPELINE

The detection environment was deployed on Microsoft Sentinel, managed through the unified Microsoft Defender portal, with a single Log Analytics workspace as the central telemetry destination. Host

telemetry was onboarded using the Azure Arc to Azure Monitor Agent to Data Collection Rule pattern, applied to both Linux and Windows endpoints.

SIEM: Microsoft Sentinel (Unified Defender Portal)

Workspace: law-soc-lab, Azure subscription, East US

Linux telemetry: Ubuntu auth/authpriv Syslog via Arc to AMA to DCR

Windows telemetry: PowerShell Operational (4104) via Arc to AMA to DCR

Attacker: Kali Linux (192.168.64.15)

Query language: KQL (Kusto Query Language)

Pipeline note: telemetry availability shaped both hunts. The Linux Syslog and Windows PowerShell script block tables ingested as expected, while the Windows SecurityEvent table (4688) proved gated on this tenant, a constraint that was documented and worked around rather than allowed to block the investigation.

3. INVESTIGATION A: SSH BRUTE-FORCE COMPROMISE (T1110)

Incident Overview

Incident type: Credential access via SSH brute-force (successful compromise)

Severity: High, confirmed account compromise

Target host: Ubuntu Linux server, OpenSSH (TCP/22)

Targeted account: mary

Source IP: 192.168.64.15

Detection source: Linux auth/authpriv Syslog in Microsoft Sentinel

Outcome: Detected, confirmed, contained at host firewall (UFW)

Methodology

A baseline confirmed that identity sign-in telemetry was unavailable, directing the hunt to host-level Linux authentication logs. The Ubuntu host was onboarded to Azure Arc, the Azure Monitor Agent deployed, and a Data Collection Rule scoped to the auth and authpriv facilities. A Hydra brute-force was then executed from the attacker host, and the resulting events were hunted in the Syslog table.

Detection Logic

Failed authentications were aggregated by source IP and account to surface the password-guessing signature, then failures and successes were correlated per source to confirm compromise:

Syslog

| *where SyslogMessage has "Failed password"*

| *summarize Failures = count() by SourceIP, TargetUser*

| *sort by Failures desc*

The decisive step correlated the 88 failed attempts with 8 successful logins from the same source, elevating the event from an attempted attack to a confirmed breach. A chronological timeline showed the activity compressed into roughly 75 seconds, consistent with an automated tool.

Response

Contained the attacker by inserting a UFW deny rule for 192.168.64.15 above the permissive SSH rule.
Recommended credential reset for the compromised account and review for post-compromise activity.
Recommended key-based authentication, fail2ban, and SSH source restrictions as preventative controls.

4. INVESTIGATION B: MALICIOUS POWERSHELL EXECUTION (T1059.001)

Incident Overview

Incident type: Malicious PowerShell execution (obfuscation, download cradle, policy bypass)
Severity: High
Affected host: JAMES-VM (Windows 11)
Detection source: PowerShell Script Block Logging (Event ID 4104)
Techniques: Encoded command, download cradle, execution-policy bypass
Outcome: Closed, true positive (simulated), all three techniques detected

Methodology

The Windows endpoint was onboarded via Azure Arc and the Azure Monitor Agent, with a Data Collection Rule using custom XPath queries scoped to the relevant event IDs. Audit process creation, command-line inclusion, and PowerShell script block logging were enabled host-side. When the SecurityEvent table (4688) proved gated on this tenant, the hunt was pivoted entirely onto script block logging (4104), which proved the stronger source.

Detection Logic

```
Event
| where EventID == 4104
| where EventData has_any ("DownloadString","IEX","FromBase64String","Net.WebClient","Bypass")
| project TimeGenerated, Computer, EventData
| sort by TimeGenerated asc
```

All three techniques were detected. Critically, the base64-encoded command, never typed in plaintext, was recovered in decoded form, confirming that script block logging defeats EncodedCommand obfuscation. Hidden-window execution likewise failed to evade logging.

Response

Confirmed true positives across all three simulated techniques.
Documented the SecurityEvent (4688) gating constraint and the Windows Security Events connector as the remediation path.
Recommended retaining script block logging as a primary PowerShell detection source and alerting on EncodedCommand, DownloadString, and ExecutionPolicy Bypass patterns.

5. CONSOLIDATED IOCs

Investigation A, Source IP: 192.168.64.15
Investigation A, Account: mary (88 failed, 8 successful logins)
Investigation A, Service: OpenSSH TCP/22
Investigation B, Process: powershell.exe with EncodedCommand
Investigation B, Behaviour: IEX (New-Object Net.WebClient).DownloadString(...)
Investigation B, Flag: WindowStyle Hidden, ExecutionPolicy Bypass
Investigation B, Host: JAMES-VM

6. MITRE ATT&CK COVERAGE

Credential Access, Brute Force Password Guessing, T1110.001
Credential Access, Brute Force, T1110
Initial Access, Valid Accounts, T1078
Execution, Command and Scripting Interpreter PowerShell, T1059.001
Command and Control, Ingress Tool Transfer, T1105

7. CAPABILITY DEMONSTRATED

The two investigations collectively evidence the core competencies of a Tier 1 SOC analyst, mapped against typical role requirements.

SIEM pipeline engineering: Built Arc to AMA to DCR ingestion for Linux Syslog and Windows event logs.
KQL and log analysis: Filtering, aggregation, correlation, and timelining across two data sources.
Detection engineering: Authored a scheduled analytics rule and indicator-based hunt queries.
Threat hunting: Proactive hunts confirming compromise on both Linux and Windows.
Incident response: Detect, confirm, contain workflow, including a live firewall containment.
Documentation: Structured reporting with executive summary, IOCs, ATT&CK mapping, and recommendations.
Adaptability: Pivoted the Windows hunt to 4104 when 4688 telemetry proved unavailable.

8. CONCLUSION

Across seven days, a Microsoft Sentinel detection capability was built from an empty workspace into a functioning SOC environment, and used to work two confirmed investigations spanning Linux and Windows. The brute-force investigation confirmed a credential compromise by correlating failed and successful authentications and closed with containment; the PowerShell investigation detected three execution techniques through script block logging and demonstrated resilience to obfuscation. Together they show not only tool familiarity but the analytical judgement, adaptability, and communication that define effective SOC work.

Repository: github.com/WiLL75G/sentinel-soc-lab-setup, full day-by-day methodology, queries, and screenshots.